Daily work report

Project: Queen Mira International School ERP

Area: Phase 1 — Finance foundation

Prepared for: Manager update

New days go at the top. Older days stay below so the history is in one file.

14 October 2026 (stabilisation gate) — Admin locked; PRM next

Status: Administrator panel stabilised and gated. Live matrix re-checked **26/26 PASS**, schema_check clean, DB audit + daily PDF + Archify architecture doc refreshed. **No PRM (Front Office) implementation in this step** — PRM starts only after this gate.

Done

1. **Admin stabilisation.** Re-ran `audit_admin_modules.py` (26/26). Committed DB/integration audit hardening (`schema_check` Admin tables, audit script table names, `.env.example` Vite flags, `DATABASE_INTEGRATION_AUDIT.md`). Word status helpers use a scored Admin-table finder (any 2+ distinctive modules) plus HTML escaping in the proof script.
2. **Daily working report PDF.** Markdown source remains `reports/daily-work-report.md`. Manager PDF: `erp complete documentation/QMIS-ERP-Daily-Work-Report.pdf` (built by `make_daily_work_report_pdf.py`). Nothing extra under `reports/`.
3. **Archify architecture documented.** Showcase overview at `.archify/architecture-project-overview/qmis-erp-overview.html` (finalize PASS). Narrative: `backend/docs/ARCHITECTURE_OVERVIEW.md`.
4. **PRM hand-off (plan only).** First PRM slice after the gate: Front Office Admission Enquiry → shared `/admissions` APIs (ensure `VITE_USE_API_ADMISSIONS=true`), then Admission List / Student DB reuse, then PRM-only screens (gate pass, re-enrollment, transfer) one module at a time.

Open / next (PRM — not started)

• Begin PRM with Admission Enquiry FE→API smoke against existing Admin admissions backend.

• Keep Admin known limitations as backlog (Roles read-only, Documents list Delete stub, Attendance read-only, Activities/TC/Communication UI click-through).

14 October 2026 (late evening) — DB / integration audit

Status: Full database, configuration and Admin data-flow audit completed. PostgreSQL `qmis_erp` matches SQLAlchemy models (47 tables); Alembic **head = current =** `0010_communication`. Live Admin matrix **26/26 PASS**; Calendar create verified in API **and** SQL; teacher dashboard **403**; pytest **400 passed**; frontend build **PASS**. Decision: **READY WITH KNOWN LIMITATIONS** (`backend/docs/DATABASE_INTEGRATION_AUDIT.md`).

Done

1. **Schema/config inspection.** Connectivity OK; no model↔DB drift; `create_all` + Alembic dual path documented; secrets redacted in the inventory.

2. **Ops hardening.** `schema_check` now warns when an Alembic revision expects Admin tables (`0005`–`0010`) that are missing; audit script table names corrected; root `.env.example` lists Admin `VITE_USE_API_*` flags.

3. **Deliverable.** `backend/docs/DATABASE_INTEGRATION_AUDIT.md` with per-module FE→API / API→DB / persist / authz / browser scorecard and setup commands.

Open

- Browser re-click for Activities / TC / Communication UI still NOT RUN this pass (live API covered).
- Admissions FE flag still defaults off unless `.env` sets `VITE_USE_API_ADMISSIONS=true`.

14 October 2026 (evening)

Status: The remaining Administrator modules now persist on the server: Documents (`cd2ec2f`), Notifications and Announcements (`d745b0d`), Calendar, TC Request Approval, Activities, Communication and the Dashboard. Migrations `0005`–`0010` are on the development database. Full backend suite **400 passed**. End-to-end Administrator audit: live matrix **26/26 PASS**, build PASS, decision **READY WITH KNOWN LIMITATIONS** (`backend/docs/ADMINISTRATOR_MODULE_AUDIT.md`). Role-module Word sheet: Admin **26 Completed / 0 Partial**. Archify project overview regenerated to include Documents, Announcements, Calendar, TC, Activities, Communication and Admin dashboard.

Done this evening

1. **Documents (committed `cd2ec2f`).** Student and employee folders on `/api/v1/documents`, authorised download, 5 MB PDF/JPG/PNG with signature checks. Browser: add/replace student files, employee upload and status change, delete folders. Teacher denied; HR employees-only.

2. **Notifications page.** Admin list reads `GET /notifications` and marks a card read when it is opened. Sample cards stay only when there is no backend session.

3. **Announcements (`0006`).** Send, list, view, acknowledge, reply, attachment. Sender and audience are server-set. Browser: list shows the two notices that exist on this campus (`AN-0001`, `AN-0002`).

4. **Calendar (`0007`).** 17 seeded 2026–27 items plus activity overlays. Browser: after load, Upcoming shows Annual Sports Day, Diwali and Unit Test.

5. **TC requests (`0008`).** File / forward / approve / reject. Admin approval screen reads the server. Live API create→forward→approve verified; browser approve click not re-run.

6. **Activities (`0009`).** Cultural / sports / competition CRUD, chess players, MD approval. Overlay onto the calendar. Live API create/approve verified.

7. **Communication (`0010`).** Real two-person inbox from campus logins. Attachments follow the same file rules. Live API start/send verified.

8. **Dashboard.** `GET /api/v1/admin/dashboard`. Browser as Administrator: Students 10, Employees 17, Leave 1, Escalations 0, TC 0, Unread 28, Announcements this month 2.

9. **Administrator audit + status sheet.** Live matrix 26/26; Word sheet coloured by status; totals recalculated (TOTAL 138 Completed / 47 Partial / 293 Not Completed / 2 Gap across all role panels).

10. **Archify overview.** `.archify/architecture-project-overview/` rebuilt (showcase finalize PASS) so the diagram lists the new Admin API modules as `mod_admin_campus` (Documents, Announcements,

Calendar, TC, Activities, Communication, Dashboard).

Not done / open

• Browser UI click-through for Activities add, TC approve button, and Communication compose was not re-run this evening (live API covers those writes). Teacher/HR denial in the browser not re-run (live API 403 PASS).

• Roles stay read-only (documented scope). Documents list Delete remains a Super Admin request stub.

• Other role panels (Teacher/Parent/…) were not re-audited this evening.

14 October 2026

Status: Class Details now stores capacity, room and class teacher on the server, per academic year, and the Administrator can edit, deactivate and delete classes from the screen. Backend, migration `0004` (applied to the development database after a backup and a rehearsal on a copy), frontend and tests are done; checked in a browser as Administrator. Nothing is pushed or merged.

Done

1. **Class Details backend (as decided at plan review).** Section capacity, room number and class teacher (stored as the login's user id, never as text); a class belongs to one academic year and is unique per campus, year and name at database level; classes are deactivated rather than deleted, and can be deleted permanently only when no student, admission, enquiry, exam, lesson plan, attendance record or mark batch uses them (the server answers with what uses it). Only an active teacher or coordinator login can be class teacher. New admin-only picker `GET /masters/class-teacher-options`. Alembic `0004` adds the columns, backfills a class's year only when its campus has exactly one year (otherwise it aborts and lists the classes), and its downgrade refuses to discard data.

2. **Class Details screens (API mode).** The list shows Academic Year, Capacity (derived), Class Teacher, Room and Status, a "Show deactivated" toggle and an actions menu (Edit sections, Deactivate / Reactivate, Delete, each behind a confirmation with a double-click guard and the server's message on refusal). The Add screen takes capacity, room and a teacher per section and is safe to repeat. A new Edit dialog changes capacity, room, teacher and active state per section and can add a section, saving row by row. Local (browser-only) mode is unchanged.

3. **Browser check as Administrator, plus local mode.** Create, edit with a new section, deactivate, show and reactivate, a refused delete of a class in use (Grade 1: 7 students, 7 admissions, 8 enquiries, 3 attendance records), a delete of an unused class, a duplicate save, and the flag-off regression (0 backend calls) all passed. **Two defects were found and fixed:** a deactivated class still stayed in the flat class and section lists that other screens' dropdowns read (my own earlier note said it did not), and the capacity cell showed placeholder dashes when nothing was set.

4. **Tests and documents.** Full backend suite **314 passed** (264 before); build and lint on the changed files are clean. `backend/docs/FRONTEND_INTEGRATION.md` now describes the new routes, rules, migration `0004` and the mirror behaviour. Details in `reports/testing-report.md`.

Not done / open

• **Not run in a browser:** Teacher or Principal on these screens (proven by pytest only), the same class in a second academic year (only one year exists in the development database), a partial section failure on screen, a real outage.

• **Open design points:** capacity is not checked against enrolment; the delete check matches class names across all years and cannot be made race-proof without a foreign key; if the unpushed Phase 2 line is restored, `TeacherAssignment.class_teacher_of` overlaps `class_teacher_user_id`.

• **A commit made by the repository owner (`10c239e`, "backend chnage") swept in this work, including the review copy of the status sheet.** That sheet is **stale** (it says no Ops API exists and no backend for announcements, calendar, communication or notifications, and the Teacher rows were not checked against `academicsApi.js`); it must be redone before anyone uses it. The Archify overview was **not** regenerated for this step.

• **Next:** redo the role-module status sheet and regenerate the Archify overview; then Attendance editing, Leave, Roles, and the missing Activities, Documents, Announcement, TC Approval, Communication, Calendar and Notifications modules, one at a time. Reviewer decisions still open: the Escalation mirror, the HR / Finance cache, User Creation status, and the Roles-editing design. The remote `main` rewind still needs the owner's confirmation.

13 October 2026

Status: Backend modules Y-0 to Y-5 built; leave, notifications, escalations and task management are wired to the frontend and checked end to end in a browser. P0 security hardening done on a review branch (not pushed); see item 18. Administrator modules started: RBAC User Creation and Roles (item 19) and Employee Database from HR (item 20) built and checked in a browser, on their own branch, not pushed. The `hr_employees.user_id` schema fix is done as a versioned migration (item 21), and so is the `school_classes.academic_year_id` fix (item 22). The Administrator's Class Details screens now use the masters API (item 23), and the Administrator's student and employee Attendance pages read real attendance (item 24). The earlier work is merged to `main` (PR #1 and PR #2); item 24 is committed on `feature/admin-attendance` (not pushed). A role-module audit of seven panels and a review copy of the status sheet are done (item 25), and the Student Database fix found by that audit is done but uncommitted (item 26); both wait for review.

Done

1. **Audit against client docs** — enumerated the 185 live endpoints from OpenAPI. Found P0-6 (RBAC CRUD) and P0-7 (audit middleware) code exists but is not mounted; backlog corrected to PARTIAL.

2. **Y-0 Approval engine** — `approval_requests`, append-only `approval_events`, `notifications`; chains per doc (lesson plan / marks: Coordinator→Principal→Director). Rules: role must match current level, initiator cannot self-approve, denial needs a reason and ends the request (resubmit creates a new row, history kept), one pending per item.

3. **Notifications** — in-app, fan-out by user or role; list + mark read.

4. **Permissions** — `approvals.submit/decide/read`, `notifications.read` for academic staff, HR, stationery manager.

5. **Y-2 Lesson plans** (Academics doc 4.5) — monthly plan per teacher/class/subject; submit goes into the approval engine (Coordinator→Principal→Director); late flag if submitted after the 25th of the previous month; reminders from the 18th to the 25th (once per teacher per month); plan locks on submit; a denied plan is revised as a new version; teachers see approval status and events; approvers see all plans.

6. **Y-3 Exams, marks and results** (Academics doc 4.6 / 4.18) — configurable exam types (CA, RCT, Midterm, Final seeded); exam schedule created by Principal, approved by DA; marks entered per exam/subject/class as a draft (preview, editable), submit locks it; chain Coordinator→Principal→DA then published; denied batch revised as new version; validation (0..max, class match, absent flag, no duplicates); auto ranks and class-high per subject; DA-set weightage per class and exam type (must total

100, locked once an exam cycle starts) with weighted final percentage; parents see published results for their own child only.

7. **Backend ↔ frontend coordination audit** — compared every frontend API call with the live backend schema. Wired and matching: auth, admissions, parents, students, attendance, finance, HR, files. Fixed: frontend called `POST /auth/logout` which did not exist (now added and audited). Fixed: exam types now seeded from the frontend's own list. Not coordinated yet (backend only, frontend still on local demo data): masters, audit, approvals/notifications, lesson plans, marks, leave. Full table in `backend/docs/FRONTEND_INTEGRATION.md`.

8. **Y-4 Leave and permission** (Academics doc 4.9) — any staff role files a request; department head (same approver map as the frontend) then HR must approve; Permission is one day, max 2 hours; overlap check; reason required for denial; no edit/cancel; approved leave is exposed through an on-leave lookup so attendance can auto-mark Leave/Permission.

9. **Security fix** — generic `POST /approvals` can no longer open approvals for lesson plans, marks, exam schedules or leave (only their own endpoints can).

10. **Frontend integration of existing modules (leave + notifications)** — before adding more backend, the leave screens now read and write through the API (`leaveApi.js`, flag `VITE_USE_API_LEAVE`): apply (including Permission with hours), My Requests, Received, view, approve/reject (rejection needs remarks). The Received list shows a request to every approver in its chain, and the Approve/Reject box shows only to the role it is waiting on. HR had no leave screen, so one was added (HR sidebar "Staff Leave Requests"). The header bell now reads real notifications (`notificationsApi.js`): unread badge, dropdown, click to mark read, refreshes every 30 s.

11. **Manual browser check of the leave workflow (live backend + frontend)** — Teacher applied Casual Leave: success message, redirect, request LR-0001 shown as Pending, stored on the server with chain Principal → HR. Principal logged in: bell showed 1, request listed under Received, approved with remarks in the UI → moved to HR. HR logged in: bell showed 1, opened the request, approved in the UI → status Approved. Server confirmed the final state and the teacher received an "Approved: Casual Leave" notification. Bug found and fixed during the check: HR's remarks box was pre-filled with the Principal's remark. Not re-checked in the UI: the teacher's own bell dropdown after approval (confirmed through the API only) and the rejection path.

12. **Y-5 Escalations with the 24h clock** — a person raises an escalation one level up (student→teacher→coordinator→principal/director…→admin→super admin, same ladder as the frontend). The receiver has 24 hours: In Review, then Resolved/Rejected (note required). If nothing happens in 24 hours the clock moves it one level up per elapsed day, tells the old owner, the new owner and the raiser, and keeps the whole trail; earlier owners keep read access. At the top level a single overdue alert is sent. Only the person who raised it can close a resolved/rejected one. The clock runs whenever escalation/task lists are read and through an admin-only sweep endpoint (no scheduler yet).

13. **Y-5 Tasks** — staff assign a task (title, priority, due date) to another staff user; content cannot be edited or deleted; only the assignee moves Incomplete → In Progress → Completed (no reopening); assigner is told on completion; an overdue task alerts both people once. Assignee lookup endpoint added.

14. **Pytest** — 80 passed at this point (71 earlier + 9 for escalations/tasks); 82 after item 16; 90 after item 17. Frontend production build passes.

15. **Escalation Management wired to the backend and checked in a browser** — correction: earlier I had only shown the backend passing tests and a health check, which does not prove the screens are connected. Now: raise screen (`escalationApi.js`, flag `VITE_USE_API_ESCALATIONS`, recipient list from a new `GET /escalations/recipients`, optional named person), lists (raised/received, Overdue label), view screen (status update with note, History trail, raiser-only Close). Browser run against the live backend: Teacher raised "Projector in 8-A…" → ESC-2026-001 shown Pending, routed to Coordinator. Coordinator's bell showed 1 and the item was in Received. Resolving without a note was refused with the server message; with the note "Lamp replaced by IT" it became Resolved. Teacher then saw Resolved with the note and a Close button, closed it → Closed (history shows raised / resolved / closed;

Coordinator sees Closed too). 24h clock: a second escalation was raised, the clock moved forward 25 hours through the admin sweep (a simulated time, not a real wait), and the teacher's screen then showed "Routed to Principal" with "auto escalated (coordinator → principal)" in the history. Two UI defects found and fixed: the read-only form showed the viewer's own escalation target instead of the record's recipient role; a direct link briefly said "not found" before loading. Not UI-checked: the Principal's screen after auto-escalation, the Rejected path, bell text for auto-escalation, other roles (director, admin).

16. **Escalation verification, second pass (browser + automated)** — covered the scenarios left open after item 15. Rejected: no note and a blank note are refused, a real note is stored and shown in the history, the raiser sees it and closes it. Principal after a simulated 25-hour climb: sees the item as received, can act, previous owner (Coordinator) is read-only. Bell: counts match the server and clicking marks read; the Teacher's dropdown shows the auto-escalation, In Review and Rejected messages. Director (item climbed two levels) can act, and cannot see an item that never reached them; Admin reads everything but cannot change status; sweep is Admin-only. A real defect was found and fixed: **lists did not refresh after the server answered** (Director showed Received 0 while the server had 1). Cause: React Compiler ignored a refresh counter in the list's memo; the list now keeps its records in state and passed three cold-cache reloads. The two lint errors were fixed (an unused re-export in `RecipientSelect`, a synchronous state update in `ViewEscalationPage`'s effect). Two new backend tests; **82 passed**; build passes; the escalation and services files lint clean (365 problems remain elsewhere in the project, untouched). Not fixed and not checked are listed in `reports/testing-report.md` (notification click does not open the item; no Admin all-items tab; Super Admin and several other roles not driven; old local-storage mode not re-tested).

17. **Task Management wired to the backend and checked in a browser** — done in two steps, as agreed. *Step 1 (basic integration):* backend now accepts an optional remark on status changes and keeps an append-only task history (who, role, remark, time); creating a task accepts a `clientRef` so a retry after a lost response never creates a duplicate. Frontend: `taskApi.js`, flag `VITE_USE_API_TASKS`, real assignees from the server, backend `Incomplete` shown as `Pending`, one task per selected person with a clear created/failed list and a safe "Retry failed" button, assigned date and overdue set by the server, document upload hidden in API mode (backend has no attachments), local-storage mode unchanged when the flag is off. My Tasks, Assign Tasks, Add Task, user select and the status modal all read from state, not from memoized counters. *Step 2 (hierarchy):* the backend now enforces who may assign to whom (superadmin→admin, admin→director, director→principal/PRM, principal→coordinator/teacher/librarian, PRM→gatekeeper manager, gatekeeper manager→gatekeeper); anything else is 403, self-assignment stays 400, and the assignee lookup only lists the caller's direct reports. Browser checks against the live backend passed for Principal, Teacher, Director and Admin: empty form blocked, several people, partial failure and lost-response retry (no duplicate), status with remarks, overdue label, notifications, role restrictions, a forged upward assignment refused with the server's message on screen, and the flag-off (local) regression. A real defect was found and fixed: **two "Task overdue" notifications per person** when two list reads ran together; the overdue flag is now claimed atomically (regression test added). **90 passed** (82 before, +8); build passes; lint shows only 4 old errors in the task pages. Not run: Librarian / PRM / Gatekeeper screens (no users for those roles), an Admin or Director "all tasks" screen (none exists), self-assignment through the screen, Super Admin, mobile. Still open: attachments for tasks, escalation auto-climb has the same kind of race (not changed). Dev data left behind: user `teacher2@qmis.edu`, tasks TASK-0001…0007 (two backdated in the database). Details in `backend/docs/FRONTEND_INTEGRATION.md` and `reports/testing-report.md`.

18. **P0 security hardening (done before any new feature)** — no work was discarded; the 33 uncommitted files were backed up and checked, a separate review branch (`review/p0-security`) was made, and every commit is reviewed before it is made. Nothing is pushed. *Protected:* literal passwords were removed from the committed tests, docs and reports (the tests now read the seed logins from settings). *Backend fixes:* (a) the API refuses to start unless `ENVIRONMENT=development`, or the production settings are safe: no default or short JWT secret, no `OTP_DEBUG`, no `*` CORS, no demo

integrations, no default database password; (b) the OTP code is only returned in development with `OTP_DEBUG=true`, and `OTP_DEBUG` now defaults to off; (c) asking for an OTP gives the same answer for a known and an unknown email (the old "No active account uses that email" 404 is gone), and unknown requests are audited; (d) rate limiting and lockout on login, OTP request and OTP verify: 5 wrong passwords lock that email for 15 minutes, 20 failures block an IP, 3 OTP requests per 10 minutes per email, 5 wrong OTP codes lock and burn the code, answers are 429 with `Retry-After`, and every lockout is audited; guessing the current password on the change-password screen is limited the same way; (e) password rules when a password is *set* (not when logging in, so existing accounts keep working): at least 10 characters, a letter and a number, no common or repeated passwords, not containing the email name, at most 72 bytes; HR temporary passwords are now random instead of a 6-digit number; (f) demo accounts (admin, HR, teacher…) are created only in development. *Frontend fixes:* removed every hard-coded login: the HR and Finance fallback logins (they used to sign in as HR / Account Head / Admin behind the user's back), the demo-account buttons and shared demo password on the sign-in page, the pre-filled parent password in the admission screen, and an unused fake "any 6-digit OTP" login. If the API rejects a login there is no local fallback any more. Offline demo sign-in now needs a development build plus `VITE_ENABLE_DEMO_LOGINS=true`. A stored role is checked with the server on page load, and a rejected token (expired or invalid) now ends the session with "Your session has expired". *Tests:* **152 passed** (90 before, +62: 52 security tests, 10 static checks of the frontend sources). Build passes; lint on the touched files shows no new problems. *Manual browser and live API checks:* sign-in failure message, lockout message after five wrong passwords, HR sign-in and HR dashboard data, Account Head sign-in and finance, HR blocked from finance (403, no fallback), sign-out, tampered stored role, rejected token on reload and in the middle of a session: all passed. Not run in a browser: the OTP screens (there is no OTP screen), the change-password screen with a must-change account, and a real wait for token expiry. *Found, not fixed:* the local database's `hr_employees.user_id` column is `VARCHAR(36)` while the code expects a UUID, so a non-HR user opening the HR "own record" page gets a server error (500); the Finance screen keeps a previous user's load error until "Retry API load" after switching users in the same tab. *Open risks:* the real local database password is in older pushed history (rotate it), the dev seed passwords are in history too, limits are per server process (use one worker or move them to Redis), there is no first-admin setup for a production database, and OTP codes are not delivered by email or SMS yet. Details in `reports/testing-report.md`.

19. **Administrator modules, step 1: RBAC > User Creation and Roles** — the role document lists 26 Administrator modules (8 completed, 6 partial, 12 not completed). This one came first because the P0 work exposed a real break: users created on the User Creation screen were saved only in the browser (with their passwords in plain text), and since sign-in is now server-only, none of them could ever sign in. Work is on a new branch `feature/admin-user-management` (cut from `review/p0-security`, two commits, not pushed). *Backend:* new admin-only API under `/api/v1/admin`: create, list (filter by role, status, search), view, update profile/status, reset password, plus a read-only list of roles and the permissions each role holds. New permissions `users.read`, `users.write`, `rbac.read` (Admin, Super Admin and Managing Director only). Rules enforced on the server: an Administrator can create only Principal, PRM, Teacher, Coordinator, Librarian, Gate Keeper Manager and Gate Keeper (a Super Admin can also create an Administrator); anything else is refused, and a created user's role cannot be changed afterwards; an Administrator cannot change, deactivate or reset an Administrator, HR, Finance, student or parent account, or their own account here; every new or reset login gets a random temporary password and must change it at first sign-in (an admin may type one instead, but it must meet the password policy); accounts are deactivated, never deleted, and a deactivated user's open session stops working at once; passwords are never written to the audit log or the profile table, and the one-time temporary password is returned with `Cache-Control: no-store`; staff codes (TEA-0001, PRM-0001…) are issued by the server; every action is audited with the acting user. New table `user_profiles` (name, gender, mobile, optional address/qualification fields); the old, unmounted `rbac.py` (whose PATCH had no authorization) was left untouched and is not used. *Frontend:* flag `VITE_USE_API_ADMIN_USERS` (defaults on whenever API sign-in is on). The Create User screen sends to the server and shows server errors (for example the

password rule); the password box is now optional ("leave blank to generate"). After saving, the list page shows the temporary password once, with a Copy button, and wipes it from the browser history so a refresh cannot show it again. The list loads from the server, with Deactivate/Activate and Reset password per row (the old Delete is replaced by deactivate in this mode). The Roles page now shows real roles with user counts, permission counts and a "View permissions" detail, read-only. The Class Details teacher picker and the Admission Enquiry PRM picker refresh from the server, so they see API-created users even on a clean browser. The browser copy of created users never holds a password. *Tests:* **189 passed** (152 before; +32 for the admin API, +5 static checks of the frontend sources). Build passes; the changed files lint clean. *Browser run against the live backend (Administrator then the created Teacher):* empty form blocked, weak password refused with the server message, create succeeded and showed the temporary password once, a reload showed the user and no password, Deactivate/Activate/Reset password from the list, Roles page with counts and permissions, the new Teacher signed in with the temporary password and was forced to change it, then reached the Teacher dashboard; the same Teacher got 403 on the admin endpoints; the Class Details teacher dropdown listed the new Teacher with an empty local cache. *Not done / open:* editing role permissions is deliberately not built (it changes live authorization; needs your decision on which module toggles map to which backend permissions); changing a user's role or email; profile photo and document upload (the upload fields on the form are not sent); Super Admin's own Admin-user screen still uses the old browser-only store; the flag-off (browser-only) mode was not re-run in a browser; the Employee Database screen still reads the browser copy, so an API user appears there only after the User Creation page or a picker has loaded the list. Dev data left behind: user `browser.teacher@qmis.edu` (TEA-0001, now Inactive). Details in `reports/testing-report.md`.

20. **Administrator modules, step 2: Employee Database connected to the HR employee API** — same branch (`feature/admin-user-management`), not pushed. *Checked first (no code):* the HR API can list and view employees (scoped by role), create them, and save the whole collection; it cannot update or delete a single employee, and has no search or paging. The screen only needs list and view, so no new employee API or store was added. HR employees and the logins made on the User Creation screen are separate records, linked only by user id or by email. *Screen (Administrator only, server mode):* it now reads the HR employee list and the User Creation logins in parallel and shows one row per person (matched by email; the HR record wins and the login is noted on it). A login with no HR record is shown as its own row, labelled "User Creation". The detail page shows HR fields, designation, category, reporting manager and status; salary reads "Restricted (HR)"; no password is shown anywhere in this mode. It has loading, error with Retry, empty, access-denied and partial-failure states (if one source fails, the other is still shown with a warning). Local (browser-only) mode is unchanged, as are the shared Principal / Director / Super Admin employee screens. New flag `VITE_USE_API_EMPLOYEES` (defaults to the sign-in flag). *Backend fix found while checking consistency:* HR's bulk save re-derives a login's active state from the HR status, so a login an Administrator had just deactivated could be switched back on by the next HR save. Deactivating or activating a login in User Creation now keeps the linked HR record in step (Active and Inactive only; Relieved and Terminated are never overwritten) and is audited. A test proves the bug is real (it fails when the fix is removed). *Tests:* **205 passed** (189 before; +12 directory tests, +4 static checks). Build passes; changed files lint clean (one lint error that was already in the list screen was fixed on the way). *Browser (Administrator, live backend):* list of 17 rows, search, department filter, HR detail, login detail, unknown id, one-row merge for a person with both records, reload persistence, and local mode (no API calls). The error, empty, 403 and partial-failure states were checked by overriding `fetch` in the page, not by stopping the server. *Found, not fixed:* HR's "own record" page returns 500 for Teacher, Coordinator, Principal and Director because the dev database column `hr_employees.user_id` is `VARCHAR(36)` while the model expects a UUID (a pre-existing defect, needs a migration; Administrator and HR are not affected). *Not run:* deactivating through the User Creation screen and watching the HR status on screen (the endpoint was checked by script and in pytest), a real outage, local-mode detail with data, the Principal/Director/Super Admin screens, and a live HR bulk save. Dev data left behind: login `live.link@qmis.edu` (TEA-0002) and HR employee EMP-0017, both Inactive. Details in `reports/testing-report.md` and `backend/docs/FRONTEND_INTEGRATION.md`.

21. **Database fix:** `hr_employees.user_id` **is now a UUID column (versioned migration)** — done before starting Class Details, as agreed. *Why it was wrong:* the model declares a UUID link to the login, but databases that already had an HR table got the column from a startup helper as a plain `VARCHAR(36)` with no foreign key (one commit on 9 Oct added the column to the model as a UUID link and to the helper as VARCHAR; `create_all` never alters an existing table, so the two definitions disagreed). PostgreSQL accepts writing a UUID into it but refuses comparing them, so every role that is not allowed to see all employees got a 500 on the HR "own record" lookup. *Inspection first:* 17 employees, 15 with no link, 2 linked (EMP-0016, EMP-0017) to valid logins with matching emails; no blank, malformed, orphan or duplicate values; nothing references the table; no views or triggers on it. *Backup:* a full `pg_dump` of the dev database was taken before any change, outside the repository (it holds password hashes, so it is not committed) and checked readable. *Migration:* the project had no migration tool, so Alembic was added (`alembic/`, `alembic.ini`, in `requirements.txt`; the database URL comes from the app settings, no credential in the files). Revision `0001` is a no-op baseline so existing databases need no stamping; `0002` converts the column in one transaction: it locks the table, **aborts and lists the employees** (changing nothing) if any value is not a UUID or names a login that does not exist, turns blank text into NULL (counted), converts, compares every row and link with a snapshot taken first (a mismatch rolls back), then adds the missing foreign key. It can be run twice safely and has a tested downgrade. It is applied by hand (`alembic upgrade head`); startup only logs a warning when the schema is behind. The startup helper no longer creates the column as VARCHAR. *Rehearsal first:* the backup was restored into a scratch database and the real migration run there (upgrade, repeat, downgrade, bad value aborts, orphan aborts, blank and upper-case values), all passing, before it was applied to the dev database. *Result on the dev database:* employee rows, both links and all audit rows identical before and after; the only new table is `alembic_version`. `GET /hr/portal/employees` now returns 200 for Teacher, Coordinator, Principal, Director, Admin and HR (it was 500 for the first four); a teacher with a linked HR record sees only their own record; HR/login activation sync still works and a deactivated user's token stops at once. *Tests:* **218 passed**, 0 skipped (205 before, +13: 8 migration, 2 startup, 3 wiring; includes a test that reproduces the original failing query and shows it works after). Build passes; lint clean. *Not run:* the Teacher/Principal screens in a browser (API only), downgrade on the dev database itself, and restoring the backup over dev. *Found, not fixed (other features):* the same VARCHAR-vs-UUID drift on `school_classes.academic_year_id` and two `admission_enquiries` columns, and `hr_employees.employee_code` is 32 characters live against 40 in the model. Details in `reports/testing-report.md` and `backend/docs/FRONTEND_INTEGRATION.md`.

22. **Database fix:** `school_classes.academic_year_id` **is now a UUID column (second versioned migration)** — done before Class Details, as asked. *Inspection first:* the model declares a UUID link from a class to its academic year, but the live column was `VARCHAR(36)` with no foreign key — the same kind of drift as item 21 (the model was changed in one commit and the startup helper added the column as VARCHAR). It was confirmed, not assumed: filtering classes by a real academic year returned 500 (`character varying = uuid`). The dev data was 1 class (Grade 1, no year) and 1 academic year. *Backup and rehearsal:* a full `pg_dump` outside the repository (readable, holds password hashes so not committed), then the real migration was run on a restored scratch copy: upgrade, repeat, downgrade, a non-UUID value, a link to a missing year and a class linked to a year on another campus (all three abort with nothing changed), and blank / upper-case values. *Migration `0003_class_year_uuid`:* same safety rules as `0002` plus the cross-campus check; one transaction, table lock, snapshot compare, then the foreign key; repeatable; tested downgrade; applied by hand. The startup helper now adds the column as UUID and the startup check reports this drift too. *Result on the dev database:* the class, the year and the sections are identical before and after; no table changed except `alembic_version`; the year filter returns 200; the HR portal still works for Admin, HR, Teacher and Principal; the foreign key rejects an unknown year. The other drifted columns were not needed for this work and were left alone (`admission_enquiries.profile_image_file_id`, `admission_enquiries.converted_admission_id`, and the `hr_employees.employee_code` length). *Tests:* 13 new migration tests; **231 passed** at that point.

23. **Administrator modules, step 3: Class Details connected to the masters API** — same branch, not pushed. *Checked first:* the backend can list and create classes (a class is created with its first section and the current academic year by default), add sections, and list academic years; it cannot update or delete a class, and it has **no** class capacity, section capacity, class teacher or room number. Nine other screens read the same browser class catalog. No new endpoint was added. *Screens (API mode):* the list reads years, classes and sections together and shows Class, Section and Academic Year, with an **Academic Year filter that runs on the server**, loading, error with Retry, and two empty states. The Add screen takes class, academic year (default: current) and sections. Saving is safe to repeat: it finds the class by name, creates it only if missing, then adds only the missing sections (a conflict counts as already present); partial failures say so and a second Save retries only the rest; the button is disabled while saving. The teacher, capacity and room inputs, the capacity column and Edit / Delete are hidden in this mode, and each screen says why. The server's classes and sections are mirrored into the browser catalog (server wins on names, browser-only fields are kept, nothing is removed) so the other screens still offer them. New flag `VITE_USE_API_CLASSES` (defaults to the sign-in flag); local mode is unchanged. *Tests:* **246 passed** (231 before, +7 API contract tests, +8 static frontend checks); build passes; the changed files lint clean (one old unused-variable error in `Components/ExportModal.jsx`, not touched). *Browser (Administrator, live backend):* list, server-side year filter (including the empty state), create class V with sections A and B, a repeat save that added only C and reported A as already present, the year filter after the creates, the offline error state with Retry recovering (network switched off in the browser), the mirrored catalog contents, and local mode (original columns and fields, a local save, no backend requests). *Not run:* a non-admin role in a browser, a real 403 on the Add screen, a partial section failure on screen, the other screens opened with an API class, mobile. *Open:* capacity, class teacher and room number need a backend change and a decision; classes cannot be renamed or deleted through the API; other class-catalog screens see server classes only after the Class Details screens have run in that browser. Dev data left behind: class `V` (A, B, C) in 2026-27, which the API cannot delete. Details in `reports/testing-report.md` and `backend/docs/FRONTEND_INTEGRATION.md`.

24. **Administrator modules, step 4: Attendance (students and employees) connected to the attendance APIs** — read-only; committed on branch `feature/admin-attendance` (two commits, not pushed or merged). *Checked first:* students' attendance already has `GET /attendance` (date, class, section, plus a roster option that lists unmarked students) and teachers mark it; there is **no dedicated employee attendance endpoint**, only the generic HR collection `GET /hr/attendance`, whose dates are free text (`dd-mm-yyyy` in dev data, ISO also occurs). The server stores **no punch-in / punch-out times for students** and no subject-wise attendance. No new endpoint was added. *Students page (API mode):* shows the marks for the chosen date (today when the date is cleared); class and section filters run on the server; choosing a class without a section also lists students nobody has marked as "Not marked" (with a section, only marked students show, because the server ignores section when it builds the roster); a count line (Present, Absent, Late, Not marked); the two time columns become "Marked At" and "In / Out" with a one-line explanation; new Late, Holiday and Not marked badges. *Employees page (API mode):* reads HR attendance joined with the employee list for names and roles, applies the date in the browser, and when nothing is recorded for the chosen date says how many records exist on other dates and offers "Show all dates". Both pages have loading, error with Retry, and empty states, and keep their filter dropdowns when a reload fails. New flag `VITE_USE_API_ADMIN_ATTENDANCE` (defaults to the sign-in flag); local mode and the Super Admin pages are unchanged. *Tests:* **260 passed** (246 before; +5 API contract tests, +9 static frontend checks); build passes; the changed files lint clean. *Browser (Administrator, live backend, after three real marks were saved through the teacher login):* the date's list with counts, the class filter showing the 4 unmarked students as "Not marked", the section filter, a date with no marks (empty message), the offline error state followed by Retry recovering, the employee list (names and roles joined, 5 records on 22/09/2026, today's empty state with "Show all dates"), and local mode on both pages (original options, columns and demo rows). *Not run:* a Teacher or other role in the screen, a real 403, marking or correcting attendance as Administrator (not built, needs a decision on who may edit and on audit for HR records), mobile layout, a real outage. *Found:* the employee attendance dates are free text on the server (an unrecognised format cannot be matched by the date filter). Dev data

left behind: three Grade 1 student marks for 13 Oct 2026 (Present, Absent, Late). *Archify:* the project overview was regenerated (`.archify/build_project_overview.py`) with the Admin users module named, an Alembic node (revisions 0001 to 0003, applied by hand) linked to PostgreSQL, 17 API clients listed, and updated "Live now" and "Still open" cards. The four automated gates passed (validate, deliver, check, real-browser check; repository evidence checked against committed files). Visual review was **not** done: the tool recommends it because 12 lines cross in the module fan-out beside the API box. Details in `reports/testing-report.md` and `backend/docs/FRONTEND_INTEGRATION.md`.

25. **Role-module audit of Teacher, Coordinator, Director, HR and Account Head, plus a recheck of Administrator and Principal, and a review copy of the status sheet** — read-only; no application code was changed for the audit itself. *Method:* signed in with each seeded login, opened every sidebar route (30 + 30 + 24 + 35 + 22 modules, plus 26 and 21 earlier) in a running browser against the live backend, and recorded the real `/api/v1` calls and status codes with every write blocked, so this covers page loads and reads only. A page loading was not counted as complete; a module counts as API-backed only if it made its own API call and the code has a client for it. *Results:* in the order Completed / Partial / Gap / Not Completed: Teacher 2 / 3 / 1 / 24; Coordinator 2 / 1 / 1 / 26; Director 3 / 3 / 0 / 18; HR 31 / 4 / 0 / 0 (35 modules; the sheet listed 34); Account Head 1 / 20 / 0 / 1; Administrator 10 / 4 / 0 / 12; Principal 3 / 3 / 0 / 15. *The sheet was wrong in several places:* it said Lesson Plan, Lesson Plan Approval and Mark Entry were Completed (the backend routes exist, `/lesson-plans` and `/marks-batches`, but the frontend has no client and the screens make no call); it cited an Ops API and `VITE_USE_API_OPS` that do not exist in the code; the Teacher cannot mark attendance from the screen (`upsertAttendanceApi` has no caller); the Coordinator's Class Attendance and Received Leave Requests never call the API. *Defects found:* `GET /finance/state` returns 403 on every fresh load for Teacher, Coordinator, Director, Principal and HR (the finance provider is mounted at the app root for every role); after a normal sign-in the Account Head sees "Finance API ... load failed" until the page is reloaded (the one-time pull runs before sign-in); every HR page requests the same 25 collections and caches payroll data in browser storage; the Student Database requested `GET /admissions/{id}` per student (item 26). A correction to my own early note: the "x4" repeats on tasks, leave and escalations are two boxes (mine, assigned) doubled by React StrictMode in development, not duplicate requests. *Review copy:* `erp complete documentation/QMIS-ERP-Role-Module-Status.review.docx` (untracked, built from the original; the locked original is unchanged and its hash matches the backup). Totals are recomputed from the rows: 480 modules, 122 Completed, 51 Partial, 305 Not Completed, 2 Gap. The 18 panels that were not re-checked keep their statuses, and only the Ops wording in their notes was replaced. Student Database and both Attendance rows are marked "Completed (pending merge)"; User Creation is plain Completed because git shows it merged in PR #2, which differs from the instruction and is flagged for review. Leave stays Partial (it mirrors to localStorage). A section 6 in the document lists three reviewer decisions (Escalation has the same mirror as Leave; HR and Finance caches; User Creation). *Checks:* 25 structural checks on the review copy passed. *Not run:* opening the file in Word or LibreOffice (neither is installed), any write workflow in a browser, an API outage, and the panels without seeded logins. A commit `49c4568` ("backend chnage") on `feature/admin-attendance`, made by the repository owner, adds only the older `.docx.tmp` file; I left it alone. Details in `reports/testing-report.md`.

26. **Student Database: no per-student admissions lookup (fix found by the audit)** — uncommitted, on a new branch `fix/student-database-admissions-lookup` cut from `feature/admin-attendance`. *Cause:* the shared mapper in `src/Common/StudentDatabase/studentDatabaseData.js` called `getAdmissionById` for every row; in API mode that returns a Promise, which is always truthy, so it was used as the record. Result: one `GET /admissions/{id}` per student (403 for Principal and Director, who hold `students.read` but not `admissions.write`; 200 for Administrator, but wasted), and the detail page showed "—" for admission-only fields that the API had returned (father name, admission date). *Fix:* in API mode the mapper now uses the row and `admissionSnapshot` from `/students` and `/students/{id}`, ignores a Promise, and no longer writes the snapshot back. Local mode is unchanged. Permissions were not widened. *Browser re-check after the fix (Principal and Administrator):* 0 admissions requests (was 9 each), the list still shows 10 rows, and the detail page shows "Parent" and 2026-10-12 as the API returns. *Tests:* four new static checks in `test_frontend_student_database_static.py`; the full suite passes

(264). Local (browser-only) mode was not re-run in a browser.

Not integrated on purpose (needs a decision first)

• **Lesson plans** — the frontend plans are weekly and approved by the Director only, with fixed names; the document and backend use monthly plans with Coordinator → Principal → Director. Needs: your decision (weekly vs monthly) and new review screens for Coordinator and Principal.

• **Marks / exams** — the frontend has no exam-schedule screen (exam names are a fixed list); backend needs Principal-creates / Director-approves. Needs a new screen before wiring.

Still open / next build

• **Administrator modules (from the role document), proposed order after RBAC and Employee Database (both done):** the `hr_employees.user_id` and `school_classes.academic_year_id` migrations (items 21, 22), Class Details (item 23) and Attendance (item 24) are done; decide whether to fix the remaining drift (`admission_enquiries.profile_image_file_id`, `admission_enquiries.converted_admission_id`, `hr_employees.employee_code` length; each a reviewed revision) and whether the backend should store class capacity, class teacher and room number; then Announcement, Communication, Calendar, Notifications page, TC Request Approval, then Activities (4) and Documents (2, needs real upload), then the Dashboard KPIs. Each as its own branch step with tests and a browser run.

• Escalations and tasks are wired and checked (items 15–17). Task gaps: no attachments, no Admin/Director "all tasks" screen, no edit/delete/reopen (by design).

• Leave: no balances/quota yet; attendance table does not yet read the on-leave lookup (student table only); emergency substitution not built. Decision: document says HR and department head both approve; the frontend used a single approver before. Backend and the frontend leave screens now follow the document (head, then HR).

• 24h clock needs a scheduler (cron) in production; today it runs when lists are read or the sweep endpoint is called.

• Exams: marks bulk is a JSON list (no CSV/Excel upload yet); question bank / key notes not built; no notification to all users when an exam schedule is published; student self-login view not possible until student users exist.

• Lesson plans: no attachment upload UI yet (fileId accepted), reminders must be triggered (no scheduler yet), frontend not wired.

• Not yet built from docs: immutability enforcement (attendance upsert and masters DELETE conflict with doc rule), RFID/iClock, refresh/logout/role switch, student user auto-create on admission fee payment.

• Doc conflict to decide: doc says no parent login; code has parent login.

12 October 2026

Status: Phase 1 identity journey closed (P1-1…P1-5). Docs + Archify refreshed. Manual browser Pass not claimed.

Done

1. **Blank screen** — `HRRoutes` eager `EMPLOYEES[0]` before HR hydrate; lazy redirect fix.
2. **Postgres** — PG18 on `:5432`; `backend/.env` password set locally (not recorded here).

3. **Admissions 500s (why A1–A8 failed)** — enquiry_code / guardian_name / ADM clash fixes + migrate.
4. **P1-4 Select Child** — `GET /parents/me/children`; ParentChildContext + smoke PASS.
5. **Seed academics logins** — teacher / coordinator / principal / director.
6. **Docs** — `API_BACKLOG` P1-4/P1-5 DONE; `TEAM_ROADMAP` / `FRONTEND_INTEGRATION` updated.
7. **P1-5 Students UIS** — `GET /api/v1/students`, `GET /api/v1/students/{id}`; `studentsApi.js`; Student Database list/detail hydrate when admissions API on; teacher/principal/director get `students.read`.
8. **Y-1 Class attendance** — `GET/POST /api/v1/attendance`, calendar map; `attendanceApi.js`; Teacher Class Attendance list loads API when enabled; `attendance.read/write` for teacher staff roles.
9. **Archify** — overview includes `mod_students`, `mod_attendance`, `studentsApi.js`, `attendanceApi.js` (finalize gates pass). Uncommitted endpoint files cite `main.py` mount until commit.
10. **Pytest** — **50 passed**.

Still open / next build

- Phase 2 **Y-2** Lesson plans
- Manual browser Parent / Student Database / Teacher attendance Pass not claimed

10 October 2026 (Phase 1)

Status: Phase 1 Finance foundation implemented and tested. **45 pytest passed.** Persistence smoke **3/3** on `:8003`. Phase 2 not started.

Done (Phase 1 only)

1. **Finance Head / Assistant permissions** (roles preserved)

- `accounthead` = Finance Head: `finance.read` + `collect` + `write` + `approve`
- `finance_assistant`: `finance.read` + `finance.collect` only (no config write / approve)
- Seeded `finance.assistant@qmis.edu` (password comes from the local seed settings)
- Routes wired: collect/send/settle/intent → collect|write; approve/confirm/concessions → approve|write; PUT state → write only

2. **Financial year (April–March)** separate from academic year

- Table `financial_years` + `/masters/financial-years*` (list/create/get/current/patch/set-current)
- Seed `FY 2026-27` (2026-04-01 → 2027-03-31)
- Rejects non April–March ranges
- Finance snapshot `meta` includes `financialYearId/Name/Start/End` + `financialYearStartMonth=April`

3. **Fee-structure validation (PostgreSQL SoT)**

- On save: feeCategories / feeStructures / fineRules validated (ids, academicYear, class, non-negative amounts, no duplicate ids)
- Snapshot still stored in `finance_records` (existing pattern — not a rewrite)

4. **Account Head persistence verification**

- Pytest: save → new login → GET marker still present
- Live smoke: `scripts/smoke_accounthead_persistence.py` → **3/3 PASS** (save, re-login, assistant PUT denied)
- Full Chromium UI click-through: **NOT RUN** this session (no browser E2E harness); API path used as refresh/logout/login evidence

Missing information (not assumed)

- Concession percentages and approval limits (school)
- Sibling/special concession rules
- Accounting / double-entry chart mapping
- Paytm vs Razorpay final approval (Razorpay left unchanged)
- Live SMTP / WhatsApp credentials
- RFID/eSSL requirement (remains disabled)

Still open (Phase 2+)

- Core fee collection expansion, refunds, yearly increase
- Live Paytm/SMTP/WhatsApp when approved
- Ops verticals; browser E2E framework

How to verify

- Head: `accounthead@qmis.edu` (password comes from the local seed settings)
- Assistant: `finance.assistant@qmis.edu` (password comes from the local seed settings)
- `python -m pytest tests/ -q`
- `python scripts/smoke_accounthead_persistence.py http://127.0.0.1:8003`

10 October 2026 (earlier — demo integrations)

Demo Razorpay/SMTP/WhatsApp + HR↔Finance click-through; RFID off. See prior section in testing report history.